CuraGraph: Knowledge Graph-Grounded Medical Question Answering with Large Language Models

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Abstract

CuraGraph combines a medical Knowledge Graph with claim-level verification for Medical QA. ClinoKG supplies biomedical facts, SubGround retrieves a query-specific subgraph, and GraphVerify checks every generated claim. CuraLM-70B serves as the generator, so the system depends on a 70-billion-parameter model and multiple data-center GPUs per query. On PubMedQA, CuraGraph improves accuracy by about five points over MedRetGen and halves the hallucination rate. The design targets answers that must be both useful and auditable: each accepted claim is linked to an explicit fact and source. The main limitation is cost, not model capacity.

Keywords

Medical QA, Knowledge Graph grounding, PubMedQA, Hallucination Reduction

1. Introduction

Medical QA systems need broad biomedical coverage and reliable factual support. SubGround [1] retrieves structured evidence, GraphVerify [2] checks claims, and ClinoKG [3] provides the medical Knowledge Graph. MedRetGen [4] is the principal text-retrieval baseline. An early clinical system based on structured drug-interaction facts [5] motivates the requirement that evidence remain inspectable. CuraGraph combines these directions around CuraLM-70B.

2. Method

ClinoKG is indexed as a typed Knowledge Graph with provenance links to biomedical text. SubGround encodes a question and extracts a connected subgraph with a bounded hop budget. CuraLM-70B receives the question and serialized facts, drafts an answer, and exposes each sentence for GraphVerify. The verifier splits the answer into claims, matches them against subgraph facts, and flags or rewrites unsupported claims. A contradiction handler marks conflicting sources rather than choosing silently. Because the generator remains CuraLM-70B, inference uses multiple data-center GPUs per query. The pipeline reports accuracy, hallucination rate, support coverage, and verification latency on PubMedQA.

3. Experiments

CuraGraph reaches 76.4% PubMedQA accuracy versus 71.3% for MedRetGen, a 5.1-point gain. Hallucination rate falls from 16.8% to 8.1%. Support coverage reaches 93.2%, while median verification adds 310 ms. The system requires multiple data-center GPUs per query, making cost and access the key deployment limitation.

System	PubMedQA accuracy	Hallucination rate
MedRetGen	71.3%	16.8%
CuraGraph	76.4%	8.1%

4. Conclusion

CuraGraph shows that ClinoKG, SubGround, and GraphVerify can improve CuraLM-70B on PubMedQA while reducing unsupported claims. The result is a strong but expensive Medical QA system: its reliance on a 70-billion-parameter model and multiple data-center GPUs remains the clearest barrier to practical deployment.

References

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